Tactix: Sweep Pass-State Checkpoint

Report to Andrew

25 September 2026

Outcome

The current N4 Sweep candidate now models pass completion independently of the retained seed schedule, matching a source-level Kissat/CaDiCaL distinction that our prior narrow completion-count change missed. A previously swept representative can remain scheduled after the pass completes; the next call must admit fresh seeds alongside it. The candidate stores per-external-variable pass marks and an incomplete flag, preserves the existing public Sweep action surface, and does not change proof emission. This is a candidate, not N4 acceptance. The AM427 technical report owns the detailed transition and test evidence.

Evidence

Formula-level tests exercise both the filtered-old-tail double completion and the completed-but-retained representative followed by fresh admission. Kernel compaction preserves external pass marks. On the candidate source, ten focused Release tests pass, including Sweep proofs and public API cases; four Debug ASan+UBSan tests pass. Documentation and rule/policy-boundary checks pass. The candidate Release solver SHA-256 is `c3eafb2bd382c2e1171f886f9cd9cf142dd92e3b538ce29e6accdce9597db2b8`.

The preceding narrow AM426 experiment, run as unthrottled paired Tier A array `1722402` with independently checked models/DSRs, is terminal and checker-clean. Both arms solve 199/200 (82 SAT, 117 UNSAT); every verdict, recorded rule counter and proof size matches. Its 1.5-second aggregate timing advantage is fragile and does not establish that the branch occurred on Tier A. The broader AM427 paired Tier A array `1722828` and dependent aggregate `1722829` are active on immutable binaries, 8 logical CPUs/32 GiB per task, with no throttle or node constraint. No AM427 performance conclusion is available yet. Neither run is the requested post-goal Tier D comparison.

Remaining gates and next action

Classify the AM427 Tier A model/proof and state-transition results, then run matched checked B/C if clean. Inspect individual regressions and seek affected-scope random-policy fuzz and reference trace evidence before N4 promotion. The AM401/412/414 fuzz owners are still in flight; canonical H2 no-option DSR checking, broad N1/N2, N3 competitiveness, other N4 rows and the production online-learning refactors remain open. Submit the full 5,000-second Tier D Tactix run only after those goal gates are satisfied.